PTUT — FINAL YEAR PROJECT

Weekly Progress Report · Week 11 of 12

TaxWise Freelance

Automated Tax Deductions & Invoicing Platform

Zero-Cost Cloud Deployment & CI/CD

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DOMAIN
FinTech, Personal Wealth & Micro-Finance

Source code: github.com/SohaibRamzan01/taxwise-freelance

1. Summary

TaxWise Freelance is **live on the internet** at https://taxwise-freelance.vercel.app with SSL (HTTPS). Every push to GitHub runs an automated **CI pipeline** and then **deploys automatically**. A full end-to-end test against the live site passes **14 of 14 steps**. Running cost: **$0**.

14/14
live end-to-end steps passing

~1.5 min
CI pipeline per push

$0
monthly hosting cost

Task from the plan	Status
Deploy to a live public URL (Vercel / cloud database) with SSL	Done — Vercel + Neon PostgreSQL, HTTPS
Automated GitHub Actions CI pipeline	Done — runs on every push and pull request
Verify production health checks	Done — `/api/health` OK; live smoke test 14/14

2. Live system

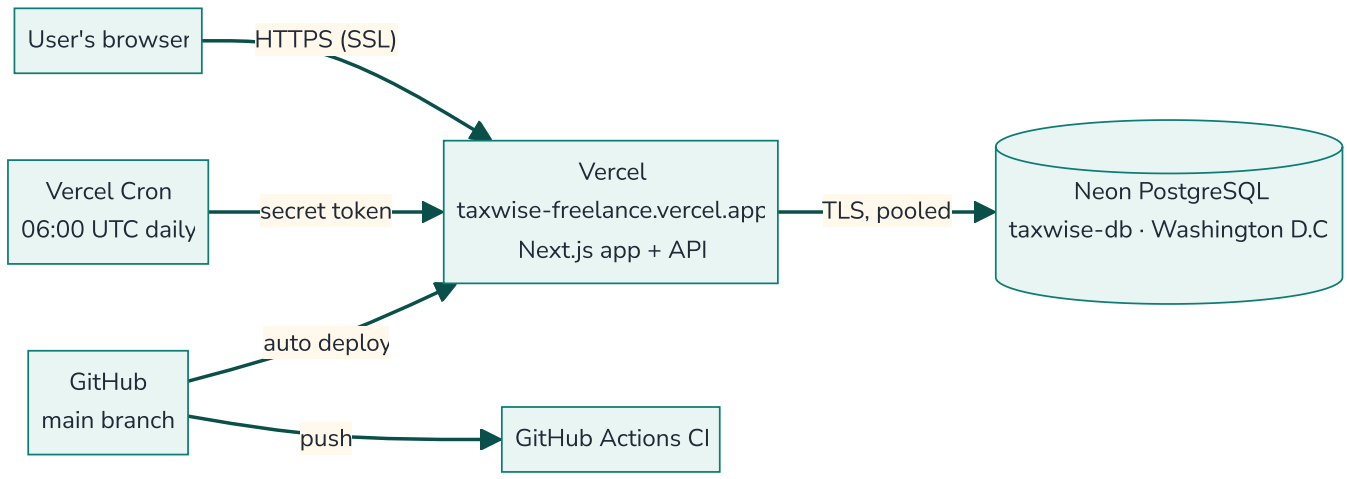


Figure 1 — Production setup

Part	Service	Plan / cost
App + API + SSL	Vercel (Hobby)	Free
Database	Neon PostgreSQL, created through Vercel's Neon integration	Free (1 GB)
CI pipeline	GitHub Actions (public repository)	Free
Daily background job	Vercel Cron	Free

Configuration: the database addresses were added automatically by the Neon integration (no passwords copied by hand). Secrets (`JWT_SECRET` , `CRON_SECRET`) are stored as encrypted Vercel environment variables, never in Git. Preview deployments use their own **database branch**, so tests never touch live data.

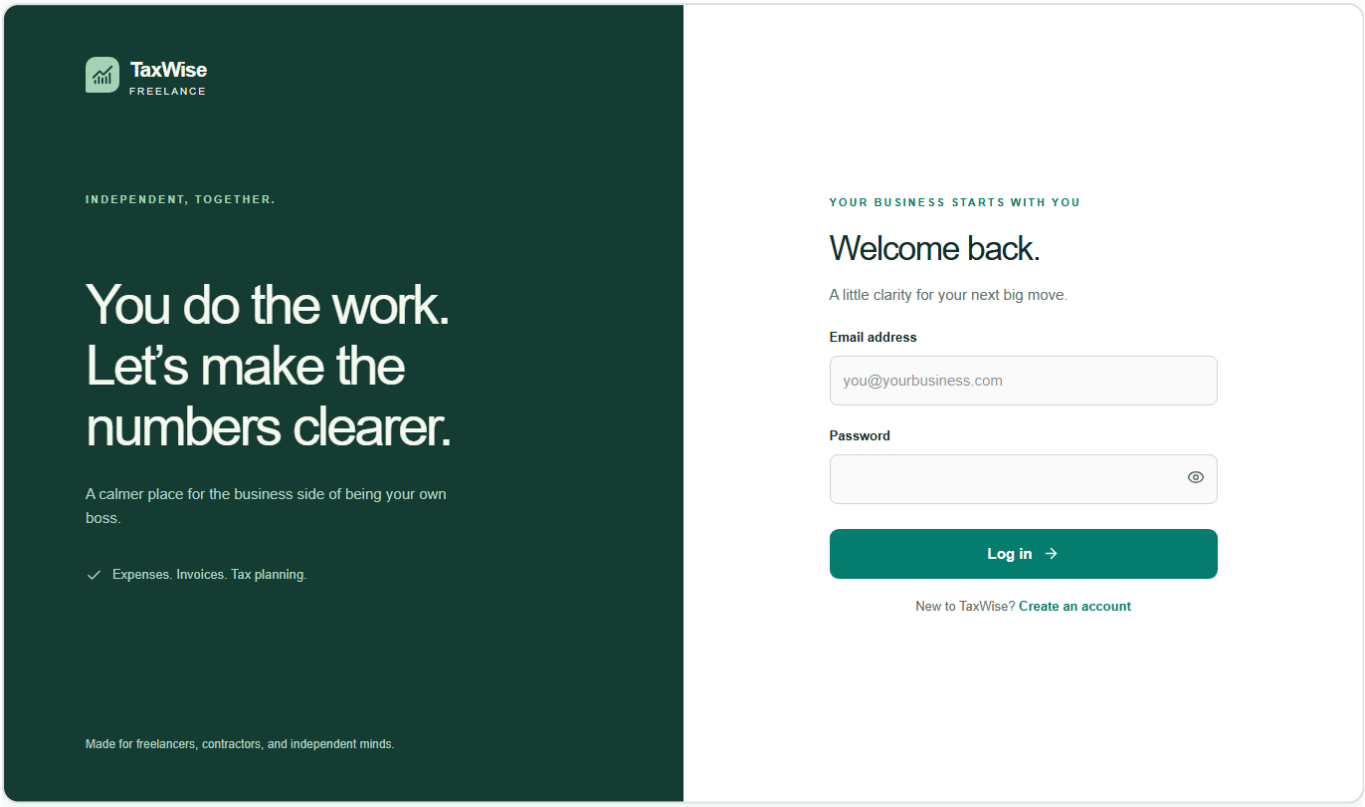


Figure 2 — The live site at https://taxwise-freelance.vercel.app

3. CI/CD pipeline

Every push to `main` runs these steps on GitHub Actions (Node 22 + a PostgreSQL 17 container):

1. Install exact dependency versions (`npm ci`)
2. Generate the database client and apply all migrations
3. Lint and TypeScript type check
4. All automated tests with the coverage gate (≥ 70%)
5. Production build
6. Bundle check — runs real OCR using only the files that will be deployed
7. Full HTTP journey test against the built app (real OCR and PDF)

If all checks pass, Vercel builds and deploys the same commit. The production build runs `prisma migrate deploy` , so the database is always up to date.

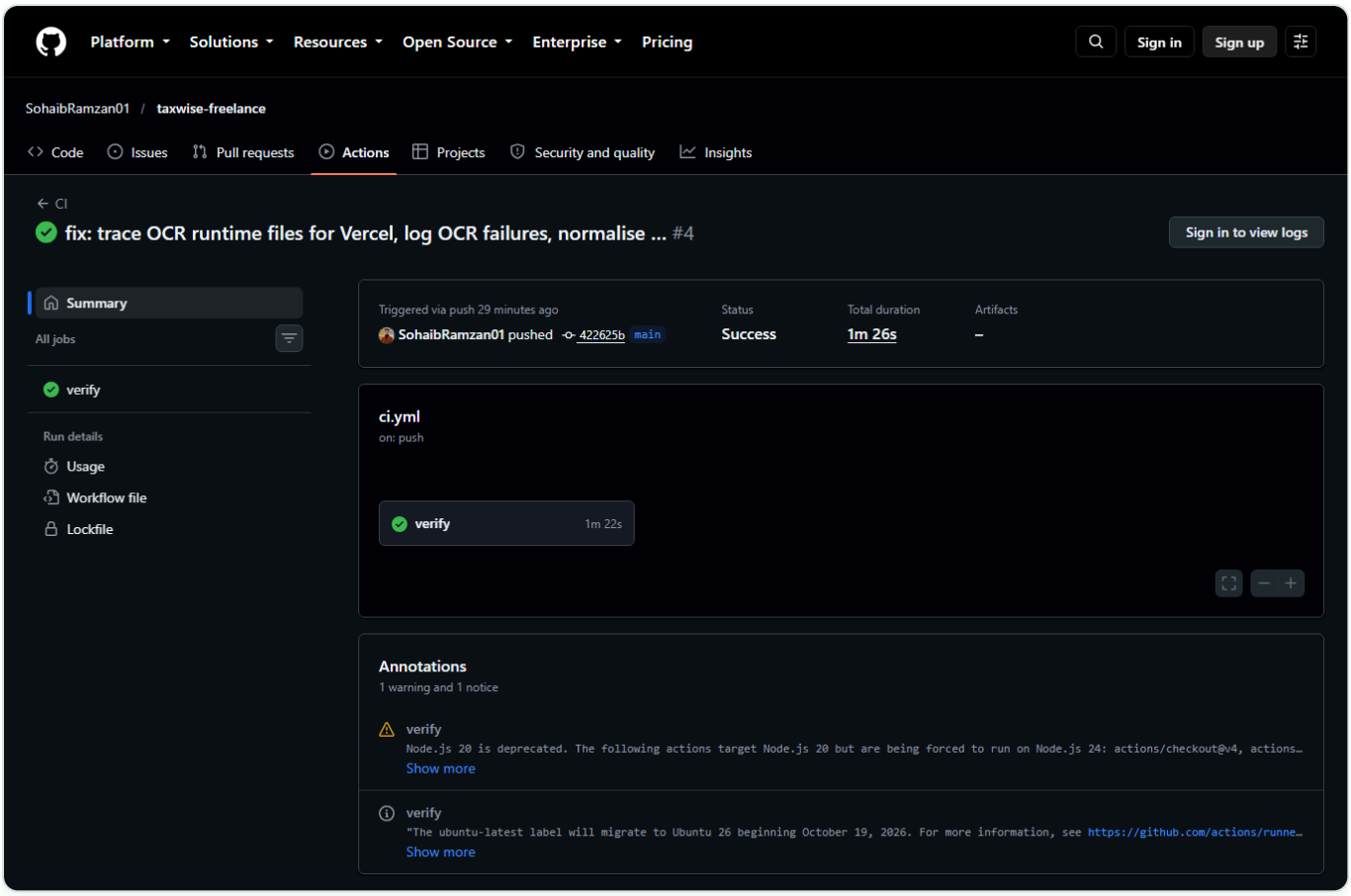


Figure 3 — A green CI run on GitHub Actions (1 min 26 s)

4. Production verification

#	Live test step	Result
1	Health check (app + database)	PASS
2–4	Sign up, create client, create invoice	PASS
5	Mark paid → exactly one income entry	PASS
6	**Scan a receipt with real Tesseract OCR**	PASS
7	Save expense, fetch private receipt photo	PASS
8–9	Tax estimate, tax-ready PDF (PDFKit)	PASS
10	Notifications; admin pages blocked for normal users	PASS
11–14	Cleanup and logout	PASS

Security headers on the live site: Strict-Transport-Security (HSTS), Content-Security-Policy, X-Frame-Options: DENY, X-Content-Type-Options: nosniff. The daily job rejects requests without its secret (401).

5. Challenges and solutions

Challenge	Solution
Receipt scanning worked locally but failed on Vercel (error 422)	Vercel uploads only the files it detects; three OCR files were missing. They are now listed in the build config, and a new check runs OCR using **only** the deployed files, so this cannot slip through again
The real OCR error was hidden	OCR failures are now logged as one structured line (no images or user data)
Database warning about SSL modes	The connection string is normalised to full certificate verification
Vercel limits uploads to 4.5 MB; phone photos are bigger	Photos are resized in the browser before upload
Preview builds could change live data	Neon creates a separate branch for preview deployments

6. Plan for next week (Week 12)

- Write the final FYP report following the academic rubric.
- Prepare the final presentation slides and the live defense demo.
- Final polish from supervisor feedback.